LESSON 6

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DIGITAL SECURITY, ETHICS AND PRIVACY

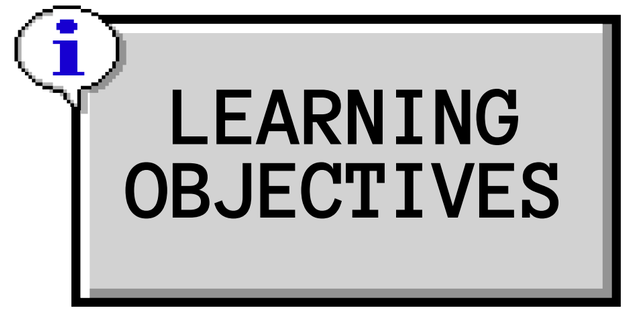
AVOIDING AND RECOGNIZING THREATHS



LESSON 6

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AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO:



- Identify risks associated with technology use
- Identify cybercrimes and criminals
- recognize issues related to information accuracy, intellectual property rights, and green computing
- Describe ways to safeguard against various types of Internet and network attacks.
- Discuss techniques to prevent unauthorized computer access and use.
- Identify risks and safeguards associated with wireless communications

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AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO:



LEARNING OBJECTIVES

- Discuss issues surrounding information privacy
- Describe how schools and businesses protect themselves
- explain the importance of inclusivity and digital access.

SECURITY

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Security is the protection from hostile forces that intends to compromise the normal state of being of something



Cybersecurity's CIA Triad

Computer Security and Cybersecurity aims to preserve these three things:

- **C :** Confidentiality of data
- **I :** Integrity of data
- **A :** Availability of data

SECURITY VS. RISK

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- Security is the state of being safe from malicious actors.
- Risk is any event or action that could cause a loss of or damage to computer or mobile device hardware, software, data, information, or processing capability.
- Risk is meant to be taken, while security is the holistic goal of every computer administrator / owner.
- i.e. We all want our computers to be secure, but there is no way to make it 100% immune from malicious actors. **How much risk are you willing to take** when you use your computer?



CYBERCRIME

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CYBERCRIME is an international breach of security often involves a deliberate act against the law.

Types of Cybercriminals:

- **Threat Actors:** A formal and umbrella term for anyone who maliciously acts upon a computer system.
- **Script Kiddies:** are individuals who want to attack computers but lack the knowledge of computers and networks needed to do so. Usually downloads pre-made hacking tools coded by other hackers.
- **Hacker:** A threat actor that breaks through a computer security without required permission

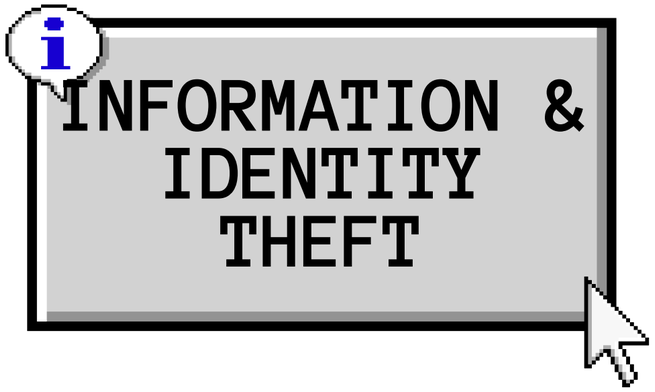
CYBERCRIME

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- **Cracker:** Someone who accesses a computer or network illegally but has the intent of destroying data, stealing information, or other malicious action.
- **Hacktivists:** a person who gains unathorized access to computer files or networks in order to further social or political ends.
- **Cyberterrorists:** Attack a nation’s national computer systems, mostly power grids and tech infrastructure that compromises national security, to instill fear and insecurity.
- **Cyberextortionist:** Threaten to expose confidential information, exploit a security flaw, or launch an attack that will compromise the organizations’s network - if they are not paid a sum of money

INFORMATION AND IDENTITY THEFT

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- Usually, attackers might want to steal information from you for personal or financial gain.
- Might use social engineering techniques to get the info they need.
- **Phishing** is a common way to trick people to entering their passwords and other info to websites controlled by threat actors.

RA 10175 Anti-Cybercrime Act of 2012

- An act Defining Cybercrime, Providing for the Prevention, investigation, Suppression and the Imposition of Penalties Therefor and for Other Purposes.

DIGITAL FORENSICS

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- Is the discovery, collection, and analysis of evidence found on computers and networks
- it involves the examination of media, programs, data and files on computers, mobile devices, servers, and networks.

INTERNET AND NETWORK ATTACKS

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MALWARE

BOTNETS

DOS

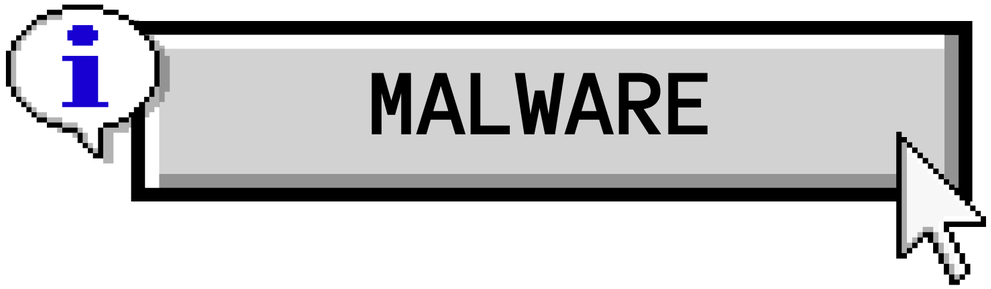
BACKDOOR

SPOOFING

SOCIAL ENGINEERING

INTERNET AND NETWORK ATTACKS

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- A contraction of the words “malicious” and “software”.
- A general term for software that are intended to disrupt a computer security

INTERNET AND NETWORK ATTACKS

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- **WannaCry** (ransomware/worm): in May 2017, WannaCry launched a massive global cyberattack.
- Impact: It affected hundred of thousands of computers across 150 countries. The global cost was estimated to be in the billions of dollars.

INTERNET AND NETWORK ATTACKS

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- also called a **zombie army.**
- is a group of compromised computers or mobile devices connected to a network, often used to attack other networks.
- usually being used in:
    - Automated Troll Farms
    - Illegal Cryto MIning
    - DDoS Attacks

INTERNET AND NETWORK ATTACKS

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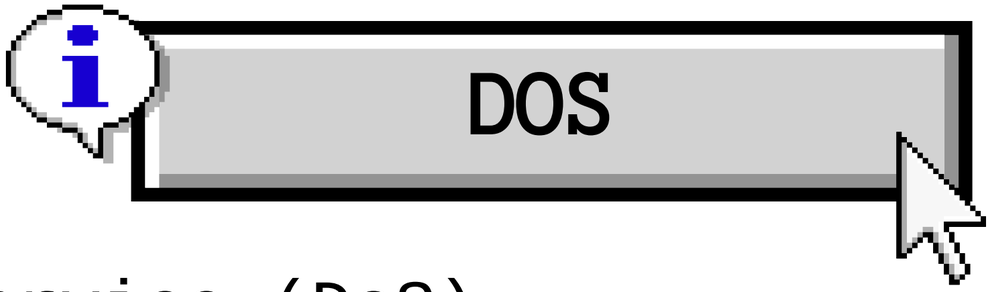


BOTNETS

- Mirai Botnet (October 2016): Targets primarily Internet of Things (IoT) devices like routers, IP cameras w/c often use default or weak factory passwords.
- Responsible for a major DDoS that temporarily brought down a significant part of the internet structure in the US.

INTERNET AND NETWORK ATTACKS

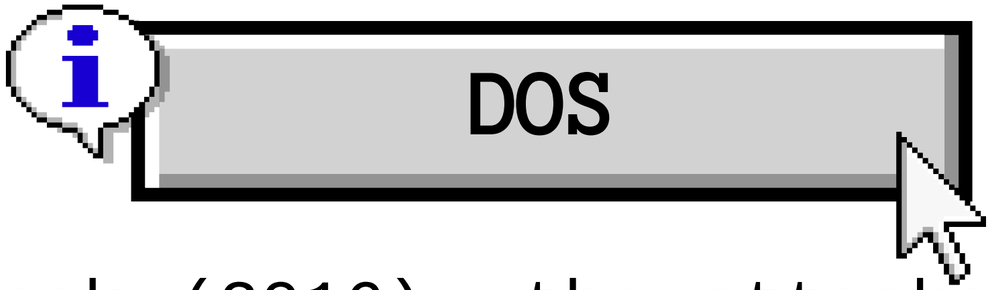
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- Denial of Service (DoS)
- a type of attack, usually on a server, that is meant to overload the server with network traffic so that it cannot provide necessary services.
- called Distributed DoS (DDoS) when executed by Botnet.

INTERNET AND NETWORK ATTACKS

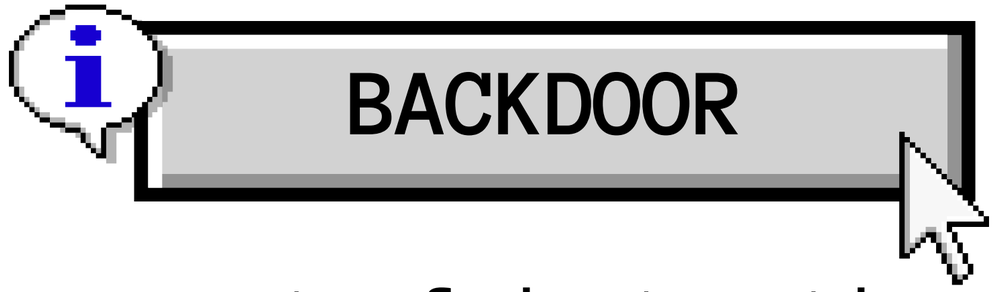
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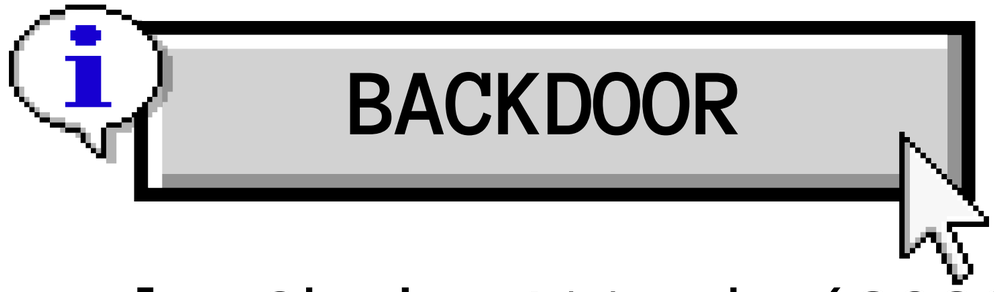
- The Dyn Attack (2016): the attackers effectively made it impossible for users to “look up” the addresses for a huge number of websites.
- This resulted in major outages for popular sites and services across the US and Europe including Twitter, Netflix, Spotify, Reddit, PayPal and the Guardian.

INTERNET AND NETWORK ATTACKS

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- is a program or set of instructions in a program that allows users to bypass security controls when accessing a program, computer, or a network.
- Backdoors are extremely dangerous because they can provide a way for hackers to launch further attacks, like installing ransomware and stealing sensitive data.



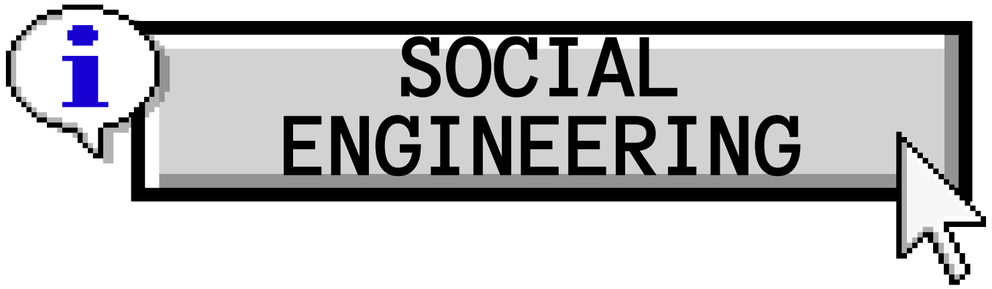
- SolarWinds Supply Chain Attack (2020) is an example of a supply chain attack where a single backdoor in one vendor’s product have the attackers a “golden key” to breach thousands of their high-profile customers.

INTERNET AND NETWORK ATTACKS

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- a technique intruders use to make their own network of internet transmission appear legitimate to a victim computer or network.
- make it look like a user uses a website safeluy, but deep inside the computer, malicious code is running that compromises the user’s network and information security.



- is a category of attacks that attempts to trick the victim into giving valuable information to the attacker.
- relies on an attacker’s clever manipulation of human nature in order to persuade the victim to provide information or take actions.

PROTECTION VS. COMPUTER ATTACKS

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ANTIVIRUS

FIREWALL

VPN
VIRTUAL
PRIVATE
NETWORKS

PROPER ACCESS
CONTROL

DATA BACKUPS

AVOID EVIL
TWINS

DIGITAL
DETOX

INFORMATION PRIVACY

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STRONG PASSWORDS

TWO-FACTOR AUTHENTICATION

BIOMETRICS

CAPTCHA

ENCRYPTIONS